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Title : Higgs boson coupling measurement at the *Large Hadron Collider* at CERN in the ATLAS experiment

The discovery of the Higgs boson in 2012 during the Run 1 of the *Large Hadron Collider* (LHC) at CERN has opened a new era for particle physics, era where the complete characterization of this new particle is of outmost importance. The Standard Model (SM) predicts with accuracy the coupling of the Higgs boson to all fundamental particles : the production and decay rates of the Higgs boson in proton-proton collisions, as well as the kinematic properties of these processes, are function of these couplings. Their measurements allow to test the compatibility of the discovered particle with the SM prediction and to measure possible deviations which could be interpreted as new physics signs.

The thesis will occur in the ATLAS experiment and will be about the Higgs boson coupling measurement in the decay mode in two photons from the proton-proton collision data collected by ATLAS at LHC at a centre-of-mass energy of 13 TeV during the period 2016-2018. The LHC restarted in June 2015 (start of Run 2), and produces collisions at an unprecedented centre-of-mass energy (1.6 times the Run 1 energy). The Run 2 should allow the ATLAS experiment to collect up to 100 fb^{-1} of data. The significant increase in energy and collected data will allow to strongly constrain the Higgs sector. Measurements in this sector are central to the Run 2 ATLAS physics program.

The decay mode in two photons was one of the two major modes for the Higgs boson discovery. But it also plays a key role in the coupling measurements because the four main production modes (gluon fusion, vector boson fusion, associated production with a vector boson and associated production with a top quark pair) can be studied in this channel.

An important part of the thesis will be devoted to the photon reconstruction and identification performance study, which are crucial elements for the considered final state. The associated systematic uncertainties will be studied using simulation and data and methods to reduce the leading ones will have to be developed. The use of a multivariate discriminant for the photon identification (not existing so far), in order to increase the fake jet rejection, will also be considered. The main part of the thesis work will be focused on the data analysis for the coupling measurement. Different analysis aspects will be scrutinized, in particular the category optimization (mainly for the vector boson fusion mode and the associated production with a vector boson), the background control in the various categories and the statistical treatment for the final result extraction. Finally, the last part of the thesis will be devoted to the result interpretation in the more general framework of the various Higgs decay channel combination. In particular, an approach based on an effective Lagrangian is being developed by the community to constrain the physics beyond the SM in the most general way. The results obtained during Run 2 will be directly used to set the first constraints in this framework.

The LAPP group members had a leading role in the Higgs boson studies in the diphoton decay channel at Run 1 and are now involved in the similar and extended Run 2 activities. The Higgs coupling measurement is clearly amongst the priority research themes of the LAPP ATLAS group for the coming years : a PhD student on this subject will have an important role inside the group. Beyond the analysis experience, from its involvement in the development, construction, testbeam and operation of the ATLAS electromagnetic calorimeter, the LAPP group has also a solid expertise in all electron and photon reconstruction and performance aspects.

The CERN proximity ($\sim 50 \text{ km}$) will allow a strong involvement of the student on the experiment site to contribute to the data taking and the collaboration meetings.